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Programming INFO Science 2201
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Visualizing NHL Data: A Statistical Exploration

Many people tend to underestimate the power and influence that data has not only in the world but especially in sports. These statistics are easy to overlook, and to draw non-data-backed conclusions based on numbers you can see on a stat sheet. The goal for this project was to dive deep into NHL statistics to try and find the real facts. The facts that can be backed with numbers and are easy to understand with well-thought-out visualizations.

One thing with sports in general is that many people can be biased based on their favorite team, player, or era, and this project aims to set aside those biases to look at true data and statistics. We will cover topics ranging from what is the best division in the 2010s era to who was the best player all time statistically. To start us off, let's look at division playoff appearances over the most recent years, from 2007 to 2022. My first two data sources were CSV files downloaded from Kaggle, stored within an NHL game data file. I first opened the playoff data frame and decided to look deeper into what these numbers I was seeing meant. I figured that for the best analysis with the clearest direction, I would use pandas to group and filter these teams into their respective divisions and analyze the data from there. Each division is made up of 8 teams, and 16 teams when you account for a specific conference (East & West). In Figure 1 below, I noticed all of the 8 teams in the Atlantic Division have made a playoff appearance since 2007. Although making the playoffs can be a great accomplishment in itself, the main goal is to win a championship. The two teams that have won a championship in this era were the Boston Bruins in 2011, reaching the full 16 wins needed. Followed by the Tampa Bay Lightning in 2020 and 2021. For any team to have a Stanley Cup champion in their division was a big deal, meaning they would be playing 26 games in total within their division. This increases competitiveness. Looking at Figure 2, we have the Metropolitan division. Same as the Atlantic division, all 8 teams have made a playoff appearance since 2007. The Metropolitan division has had 2 Stanley Cup champions: the Pittsburgh Penguins in 2009, 2016, and 2017, followed by the Washington Capitals in 2018, as seen on the graph. Further analyzing this data, I noticed the pure dominance of the Penguins, who did not miss a single playoff appearance within 15 years. When you have such a dominant team within your division, it becomes difficult when knowing you only have 3 guaranteed spots out of 8 teams. Moving on to figure 3, the Central Division, the Chicago Blackhawks ruled over this division for years from 2009 to 2017, being one of their latest playoff appearances. When looking at the graph, you can see the Chicago Blackhawks' Stanley Cup

victories in 2010, 2013, and 2015, when they became a dynasty. Followed by the St. Louis Blues with 1 in 2019 and the Colorado Avalanche with 1 in 2022. The Central Division has the most champions in the NHL during this era. Lastly, when we look at Figure 4, the Pacific Division, we see that this division is made up of 7 teams, making their odds for a playoff spot much easier. Same with the other 3 divisions; all teams have made a playoff appearance. During this era, the Los Angeles Kings were the only team to win a championship in 2012 and 2014. Starting, the Central Division has had 5 championships, the Metropolitan with 4, the Atlantic with 3, and the Pacific Division with 2. Based on these findings, I can not yet decide which division is the most competitive; however, I can look at the top teams from these divisions to further my explanation.

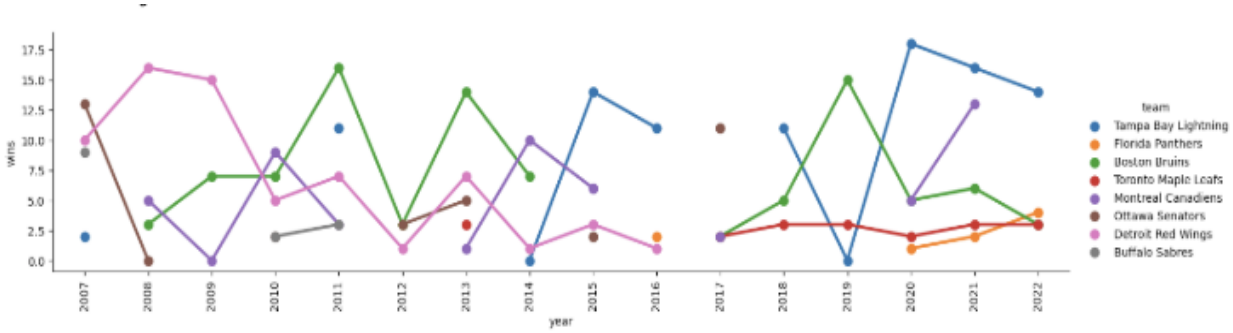


Figure 1: Total number of playoff appearances and wins from 2007 - 2022 Atlantic Division

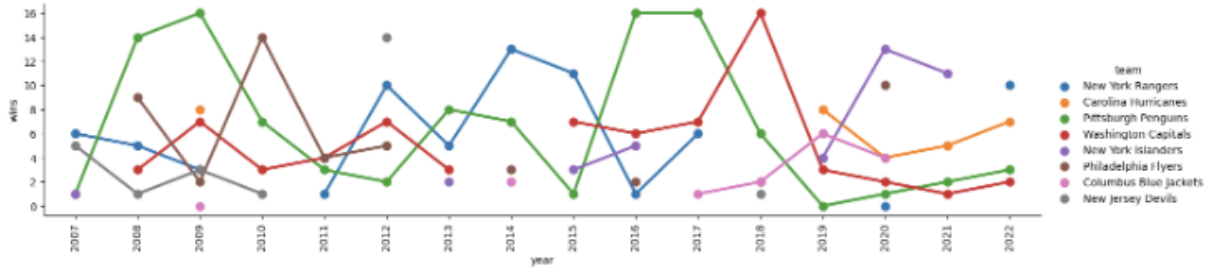


Figure 2: Total number of playoff appearances and wins from 2007 - 2022 Metropolitan Division

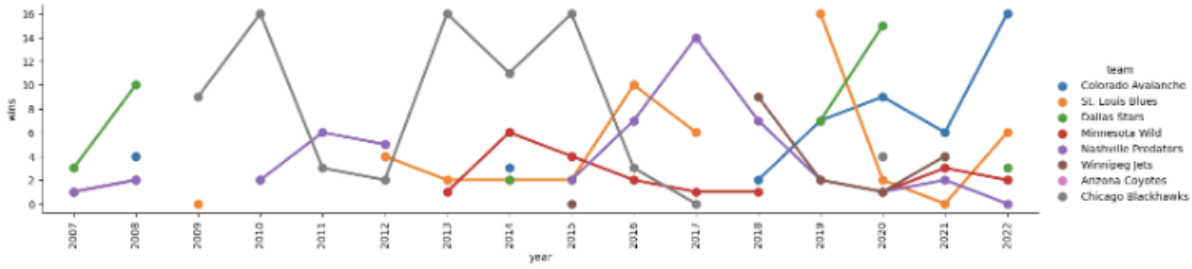


Figure 3: Total number of playoff appearances and wins from 2007 - 2022 Central Division



Figure 4: Total number of playoff appearances and wins from 2007 - 2022 Pacific Division

The next section I wished to explore was a few of the individual teams that dominated within their division. As seen in Figure 5, I have analyzed the years and the wins that the Tampa Bay Lightning have had from 2007 to 2022. We can see that they tend to make the playoffs regularly, especially from 2018 to 2022. Although they were not as dominant in the late 1990s and the early 2000s, we can draw this up to rebuild years until they finally got their hands on a championship. Moving on to Figure 6, we have the playoff wins of the Chicago Blackhawks. We can see the total superiority that they had from 2010 to 2015. Winning a single championship for any franchise could be a lifelong goal for an organization. The Blackhawks won 3 championships in 6 years, fully cementing themselves in history and as a dynasty. In Figure 7, the Pittsburgh Penguins were plotted. With this visualization, we can see that from 2007 to 2022, they have not missed a single postseason. This is a great accomplishment in its own, and you would want to begin looking into their roster. Such as players Sydney Crosby, Evgeni Malkin, and Marc-André Fleury. When you are in a division with a team that has proven to make the playoffs year over year, you begin to think you are only fighting for 2 playoff spots left. The last Figure 8 is the Los Angeles Kings. When looking at the visualization, you can see the two years when they won the Stanley Cup in 2012 and 2014. They missed the playoffs from 2003 to 2009, as well as having a gap from 2016 to 2018 and from 2018 to 2021. After examining these plots, it became hard to argue with the Central Division having the Chicago Blackhawks and the Metropolitan Division having the Pittsburgh Penguins.

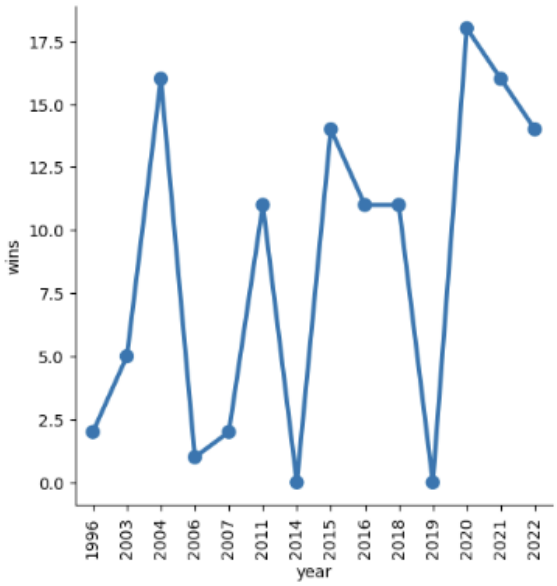


Figure 5: Tampa Bay Lightning playoff wins from 1996 to 2022

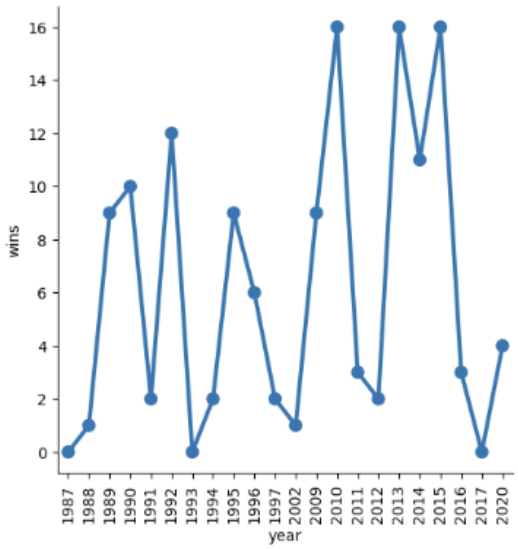


Figure 6: Chicago Blackhawks playoff wins from 1987 to 2020

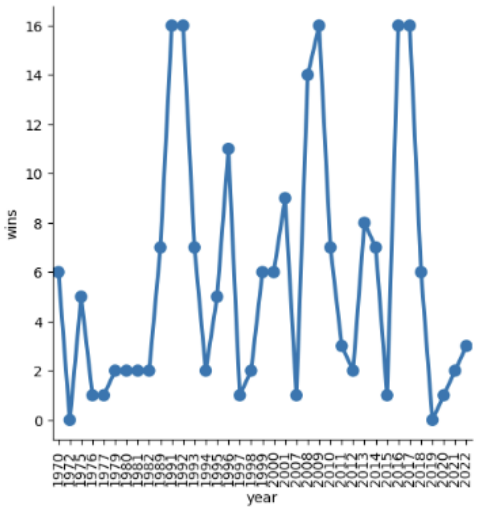


Figure 7: Pittsburgh Penguins playoff wins from 1970 to 2022

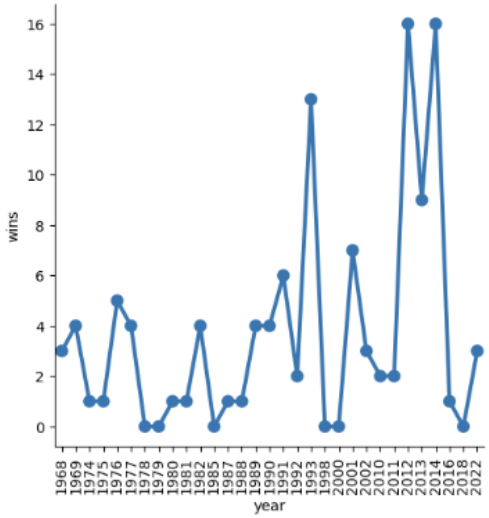


Figure 8: Los Angeles Kings playoff wins from 1968 to 2022

Following the previous findings, I was now curious to see what the divisions' overall win percentage was by team. To do this, I further looked into the playoffs data frame and was able to use the recent divisions records data frame by filtering only the records from 2007 and beyond. I will show these plots in the format of a bar plot. Starting with Figure 9, we have the Atlantic Division, the Buffalo Sabers have yet to make the playoffs after 2011, and the Detroit Red Wings used to have an above .500 win percentage, and their last playoff appearance was in 2016, where they had a .250 win percentage ¼. Following that, we can see the Lightning begin to excel, with their win percentage in the playoffs going as high as 0.720 in 2020, leading to them losing 1 in 4 games on average. In Figure 10, we have the Metropolitan Division. In this plot, we can see the

Penguins' dominance from 2008 until 2018, where their win percentage was above 0.500. As well, with the Washington Capitals, we can see that in 2018 they had a win percentage of nearly .800, in 2018, the Washington Capitals won the Stanley Cup. Furthermore, a majority of playoff teams came from the Metropolitan division. On to Figure 11: this plot is the Central Division. We can see the Blackhawks' dominance with having a playoff win percentage above .500 in the years 2009, 2010, 2014, and 2015. In 2018, we can see the Avalanche begin making playoff appearances and won the Stanley Cup in 2022 with a win percentage above 0.800, meaning they would lose on average 1 in 5 games. Lastly, in Figure 12, this plot is the Pacific Division. We can see that teams are quite equal when compared to other divisions. We can see the Kings in 2012 and 2014 having a win percentage above .600. In 2014, they won the Stanley Cup and have yet to make a big move in the playoffs as a division since.
Following this data, we can conclude that the Central Division has been the most dominant in the 2007 to 2022 era. Not only do they have 5 championships within this time, but they have had multiple teams with an above .500 win percentage. Close after, I would put the Metropolitan division just based on the Penguins' playoff supremacy.

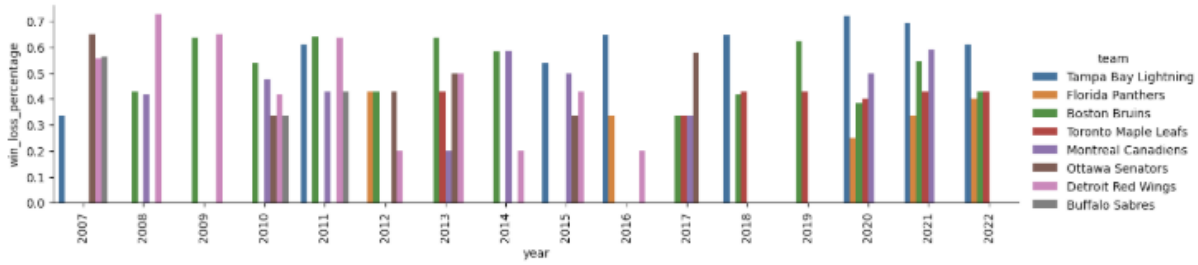


Figure 9: Atlantic division playoff win percentage by year

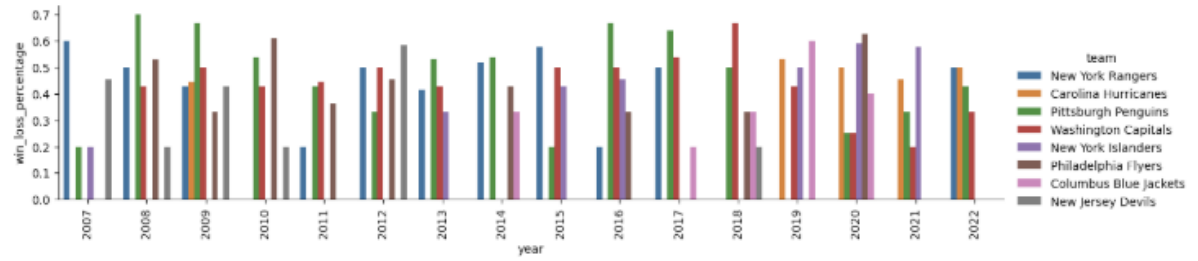


Figure 10: Metropolitan division playoff win percentage by year

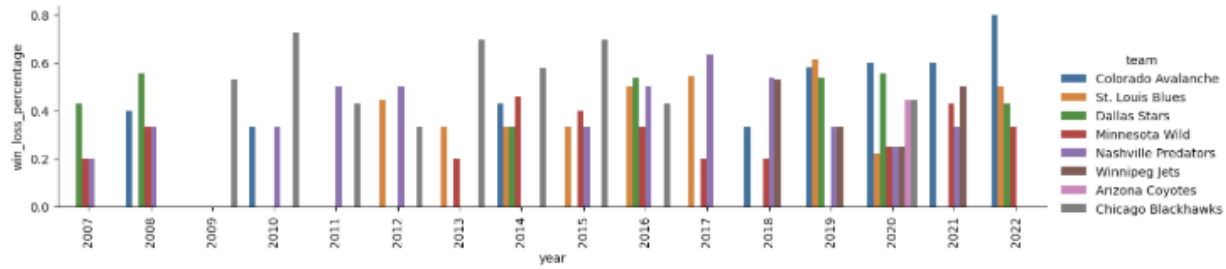


Figure 11: Central division playoff win percentage by year

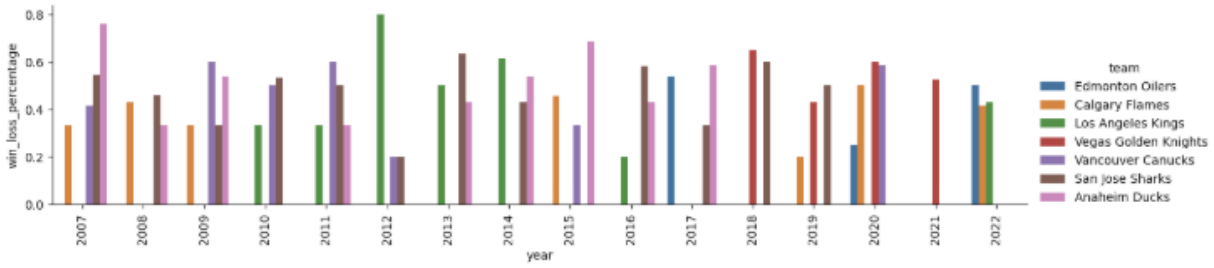


Figure 12: Pacific division playoff win percentage by year

After looking at playoff performers, I was curious to see some statistics on what teams tend to underperform in the playoffs. To do this, I will create a new named data frame while pulling data from the playoffs and game CSV files. I then proceeded to filter the teams that have scored less than 40 goals and had a win/loss % less than 0.550 in the playoffs. In Figure 13, we have a scatterplot of playoff teams' goals vs. the win%. In this visualization, there is a strong correlation between the number of goals scored and how likely you are to win. The Los Angeles Kings scored 41 goals in the playoffs and had a win percentage of 0.400. We can also see that the Nordiques, now known as the Colorado Avalanche, scored 41 goals, but had a win percentage of 0.540. Nonetheless, the data is clustered to the left, meaning the fewer goals a team scores, the more likely they are to lose a game or series.

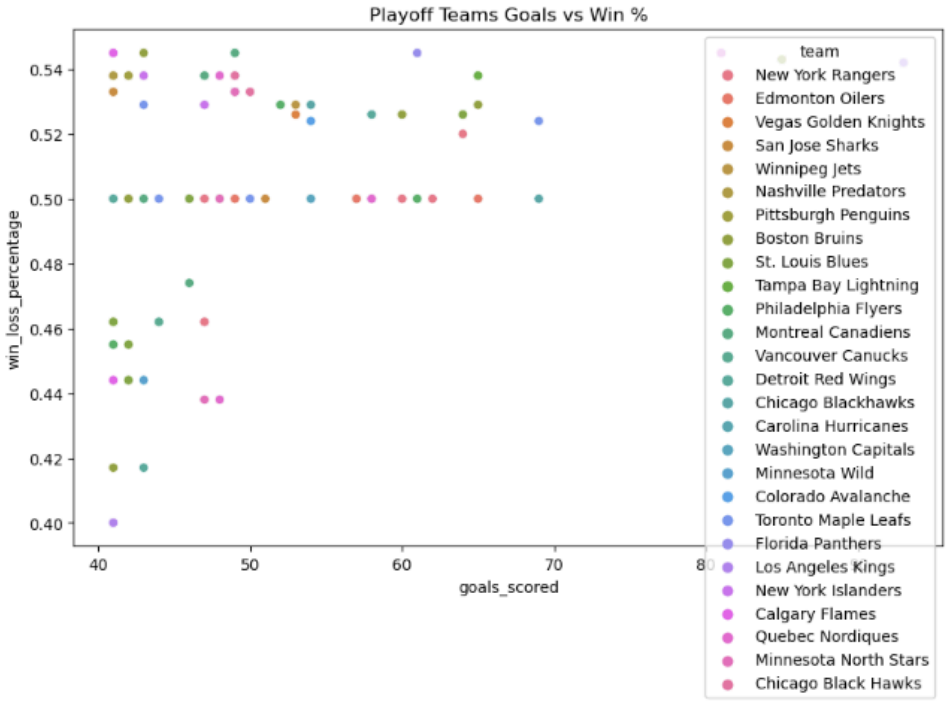


Figure 13: Playoff Teams Goals vs Win%

Using information from Figure 13, I also wanted to look at the team's goal differential in the playoffs. For Figure 14, the Winnipeg Jets have the highest goal differential on average in the playoffs of 12 goals. A majority of the teams on this plot range from 0 to 5 in goal differential, which isn't a bad spot to be in. On the opposite side, the Carolina Hurricanes have a goal differential of -10, which can explain why they have not been a great team in the playoffs since 2007. Although you can't judge offense without looking at the defense. The black bars represented in the columns show the range of uncertainty and how much variation there is in that team's playoff goal differential across seasons or games.

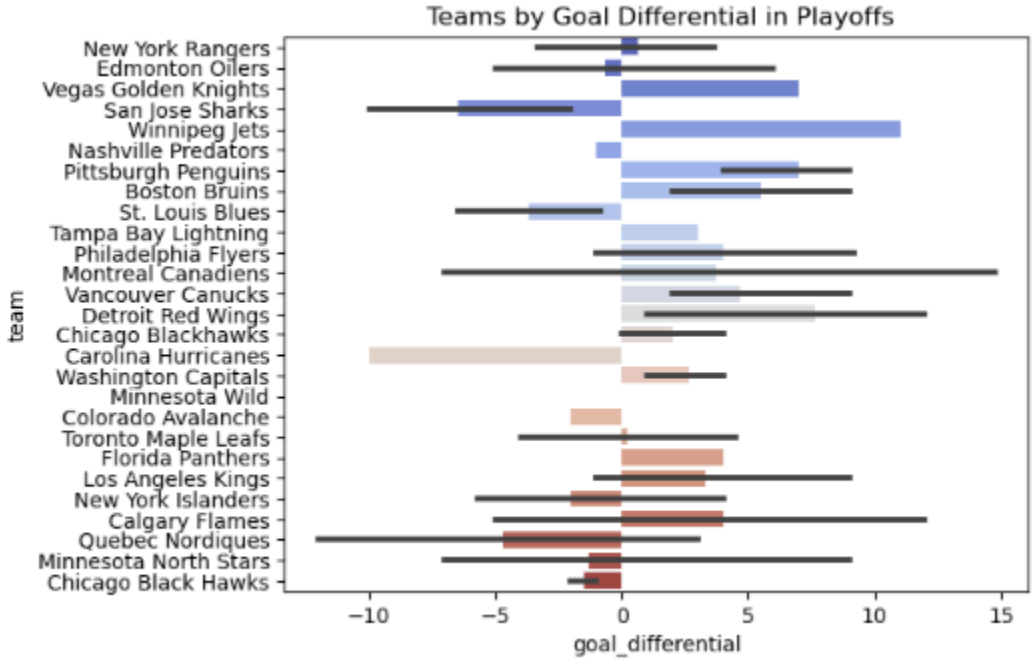


Figure 14: Teams' goal differential in playoffs

Moving away from the use of CSV files, I then went on to explore API file data. It was a compiled document with current metrics within the NHL from specific franchises, draft data, attendance, playoffs, season records, and specific milestones for each player. The API had a list of different endpoints to pull the information from the file. To start off, I wanted to look at playoff attendance, regular season attendance, and total attendance data. After looking at the total attendance data frame in descending order from greatest to least, the best attendance recorded in NHL history was the 2023 - 2024 season, with in-person attendance accounting for 24,499,541 people. The average arena has a capacity of 18,500. Looking at Figure 15, we can see how much the NHL has grown since it first began as a league. They grow in a linear trend nearly every year, despite years 1978 - 1789, 1994 - 1995, 2004 - 2005, and 2020 - 2021. Although these can be explained for multiple reasons, in 2004 - 2005 the NHL had a lockout and did not have a season at all. Similar to 2020 - 2021, the COVID-19 year, when fans were not allowed to attend games in person. It would be interesting to look into the reasons why NHL attendance is growing so

much. Could it be because of social media attracting new fans, the 4 Nations tournament, or because they keep expanding teams to new cities? Their lowest attendance season was in 1957 - 1958 with 2,444,283 people, and it has grown to nearly 25 million people in attendance for the 2023 - 2024 season.

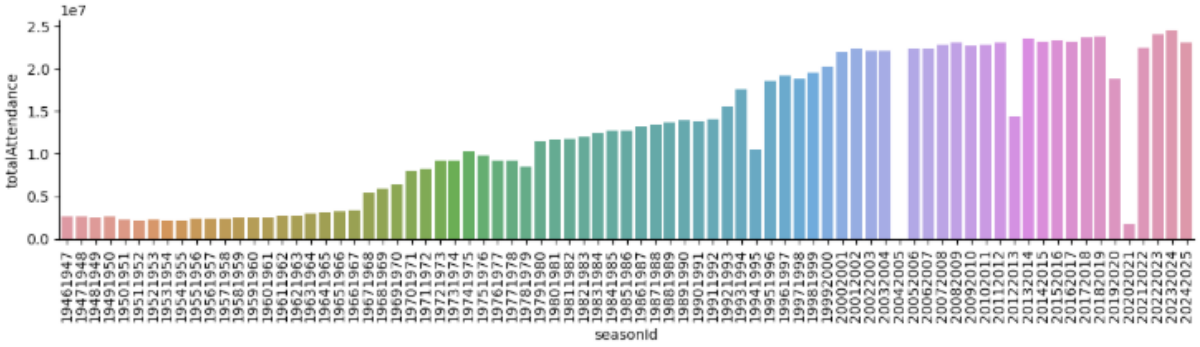


Figure 15: Total attendance from 1946 to 2024

Lastly, for my analysis of attendance records in the NHL, I was curious to look at regular season statistics from 2004 - 2024. This data was found in the attendance API under the column for the regular season. In Figure 16, we can see a line plot that shows the average attendance for the regular season from 2004 - 2024. In 2013, the NHL had a spike in attendance at 21,800,000 and then dropped until the 2016 -2017 season. Year over year, the attendance keeps growing at a rate of 123,000 per year. The latest season has had its peak for the NHL at 23,050,000. We can begin asking questions to see what the reasons for this growth are.

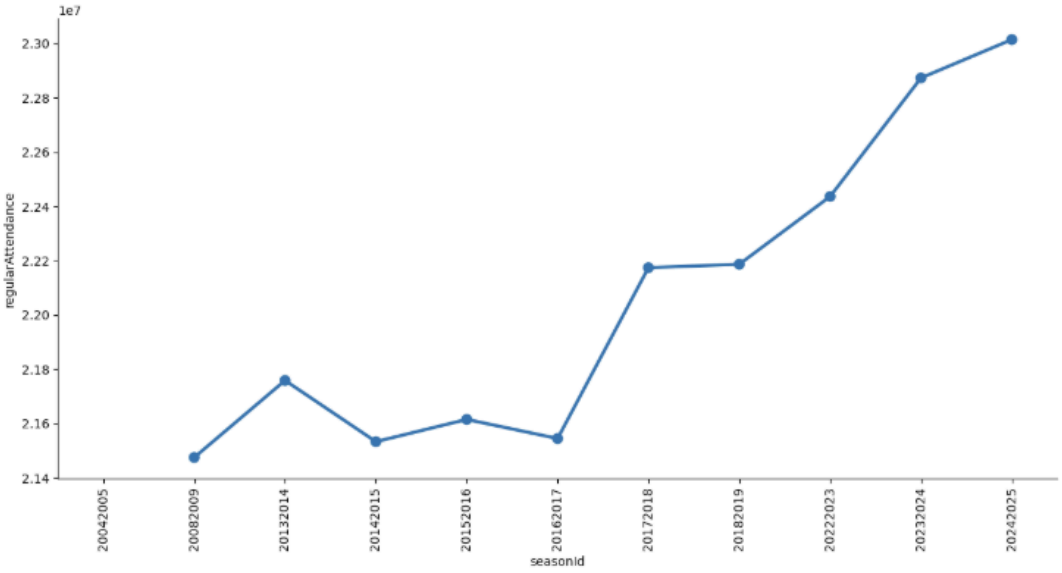


Figure 16: Regular season attendance from 2004 - 2025

Lastly, let's look at the players with the most career points in NHL history. For this, we will use a new API endpoint called milestone 1000 and create a data frame. I then filtered this data to the top 10 in career points. In Figure 17, we can see Wayne Gretzky with quite possibly the most impossible record to break in all of sports. Throughout his career, he totaled 2,857 points in 1,487 games. The next closest player is Jaromir Jagr with 1,921 in 1,733 games. It is reasonable to ask if this record will ever be broken.

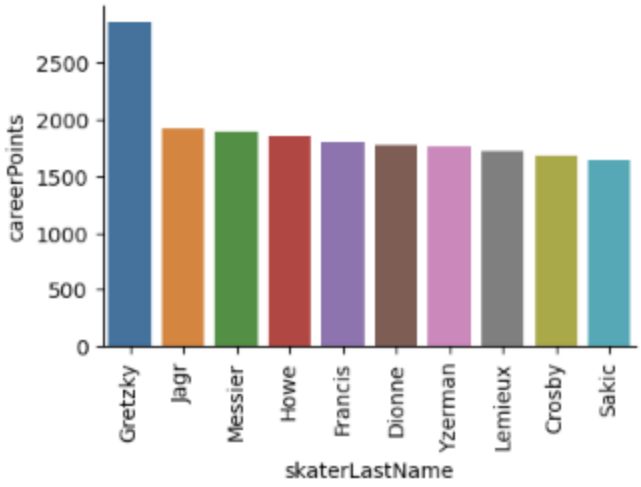


Figure 17: Total career points by player

The next steps I see in making this project better would be to put more research into the individual players for teams that excel in the league. This would allow me to ask further questions that I have as to what position matters the most: do you need goal scorers, or do you need good defense? I would also like to look into goal differential by specific teams in the regular season to see where they are falling short. I was able to convey that the Central Division is the most competitive in the NHL by looking at win percentages and overall dominance in the league. As well as being able to spot the correlation between underperforming teams and goals scored. I also looked into how large the NHL fanbase is growing, and you can see that in the plots listed. Although there are still many questions I am eager to find the answer to, will the point record ever be broken, and who is closest to that goal? What are the reasons why the NHL continues to grow each year? What do they do differently from other professional sports leagues? Lastly, what NHL division is on track to be the new dominant force within the NHL?